UNIVERSIDADE DA CORUÑA

Co-funded by the Erasmus+ Programme of the European Union

SUSTAINABLE SHIP AND SHIPPING 4.0
SEAS 4.0
Erasmus Mundus Joint Master Degree
Excellence in education for a new generation of Naval Architects and Marine Engineers

SIMULATION OF SHIPBUILDING PROCESSES (SOSP)

Report for

LEAN MANUFACTURING: CNC MACHINING WITH ERROR DETECTION

Presented By

LEMUEL HORNSBY-ODOI

Submitted To

ADOLFO LAMAS RODRIGUEZ

Table of Contents

INTRODUCTION

Lean Manufacturing has become a cornerstone philosophy in modern production systems by focusing on the pursuit of waste elimination and continuous improvement and delivering value to the customer. Core principles include Heijunka, meaning leveling production to reduce variability, and Jidoka, automation with human intelligence; machines automatically detect abnormalities and stop the process from producing further defects. These concepts improve the stability of flow and quality of products but also align with the paradigm of Industry 4.0, which is characterized by intelligent and responsive manufacturing.

In a precision CNC machining environment, it is crucial to maintain consistent throughput and ensure high-quality output. Variations in client demand, product complexity, and defect occurrence can easily disrupt flow, create excessive work-in-progress, or cause costly rework. This project applies Lean Manufacturing principles to a simulated CNC machining line developed in FlexSim. The study will be done based on three models: a Heijunka model that manages mixed client demands through leveled scheduling; a Jidoka model that integrates automated error detection and repair logic; and a combined model that merges both approaches to assess their combined effects on performance. These simulations will optimize key parameters such as pitch time, defect rates, and resource utilization to illustrate how production leveling and autonomation can jointly improve efficiency, quality, and responsiveness in CNC operations.

Theoretical Background

Lean principles waste reduction, flow, pull, perfection

Lean Manufacturing focuses on eliminating, smoothing flow across processes, using pull to produce only what downstream needs, and pursuing perfection through continuous improvement. Classic Lean identifies waste such as waiting, overproduction, defects, inventory, motion, over-processing, and transport; the system aims at reducing variability and lead times, at the same time increasing value to the customer.

Heijunka — production leveling in mixed-model environments

Heijunka levels volume and mix over time so that capacity, inventory, and labor are not whipsawed by spikes in demand. In mixed-model lines, leveling sequences products into small, repeating cycles or lots for the purpose of stabilizing upstream loading, reducing WIP, and improving responsiveness without large finished-goods. By decoupling the production cadence from volatile orders, Heijunka improves overall system flow and reliability.

Jidoka - detect abnormalities and build quality at the source

Jidoka (autonomation) is a process that enables the process to detect abnormalities and stop safely, signal for help, and fix root causes before defects flow downstream, embedding "quality at the source." Originated from Toyoda's automatic loom, Jidoka enables operators and machines to avoid defect proliferation and reduce rework and enhance safety and stability

FlexSim in Lean modeling - discrete-event simulation for performance analysis

DES is a highly applicable method for studying Lean transformations because of its power in capturing resource contention, variability, and control policies such as pull rules, leveling logic, and Jidoka stops. Using tools like FlexSim, analysts test scenarios such as Heijunka pitch, WIP caps, and defect detection policies; measure the impact of these scenarios on flow time, WIP, throughput, and utilization; and communicate this information prior to implementation

System Description and Modelling Scope

This study models a CNC manufacturing line that converts three raw material inputs—bars, pre-cast shapes, and sheet-metal blanks—into finished products for three client families. At time zero, the Warehouse contains 200 units (40 from Source 1, 60 from Source 2, and 100 from Source 3). All items follow a common processing route: Preparation → Rough Machining → CNC → Inspection → Pack/Combine → Finished Goods.

Three client product types enter this shared flow:

- Client 1: precision gear housing (longer, tighter-tolerance work)
- Client 2: propeller impellers (medium complexity)
- Client 3: shaft brackets (short, high-volume work)

The analysis compares two FlexSim models:

1. *Heijunka* – leveled production to smooth mix and workload
2. *Jidoka* – built-in quality with a Repair loop that returns defective items to the failing stage

The flowchart below summarizes the overall manufacturing process and the Jidoka defect-handling loop

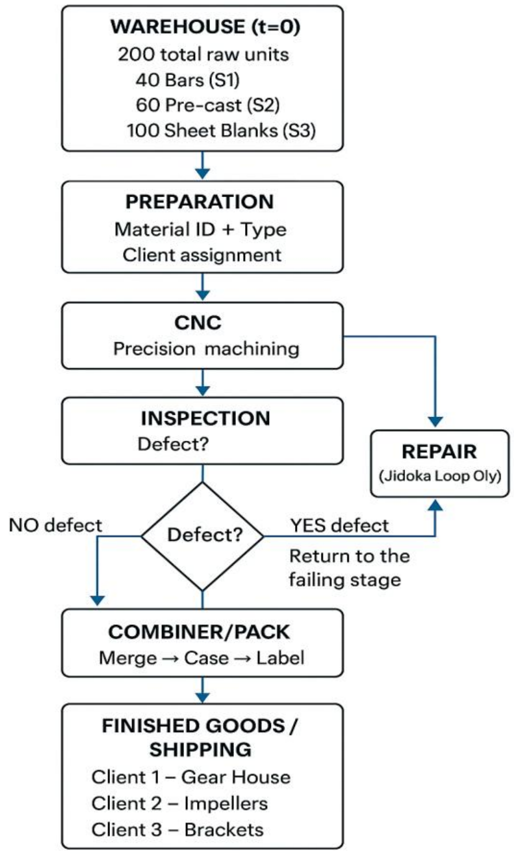


Figure 1 FlowChart of Process

METHODOLOGY

Modelling Environment

The simulation models were developed in FlexSim 25.1.3, using its discrete-event modelling capabilities to represent the flow of mixed raw materials through a shared CNC manufacturing line. All elements including sources, queues, processors, inspection stations, the repair loop, and final shipping were arranged within a common workshop layout to maintain structural consistency across experiments.

Model parameters such as processing times, defect probabilities, and buffer capacities were defined through global variables, ensuring uniform behaviour and allowing controlled scenario changes. Tracked Variables were used to measure key performance indicators: Completed Orders, Lead Time, WIP, and Makespan. Additional triggers (Set Label, Close/Open Ports, Pull Strategy customization) provided the logic required for production leveling and defect handling.

CNC Production System

All products follow a common Bill of Process: Preparation → Rough Machining → CNC → Inspect → Combiner/Pack → FG/Ship. Three upstream material sources feed the line via dedicated Preparation feeder queues with an initial warehouse stock of 200 units (40 bars, 60 pre-cast shapes, 100 sheet-metal blanks). At Preparation, a pull strategy implements Heijunka logic. Processing times are defined by client (Values-by-Case) to reflect product complexity: Client 1 (precision gear housing), Client 2 (propeller impeller), Client 3 (shaft bracket). Quality risk ("defectology") can be applied at Rough and CNC; in Jidoka scenarios the CNC includes error detection that halts the machine and returns items to the failing stage via a Repair loop (quality at the source). Finished parts are routed to client and then shipped; service is assessed against per-client demand while monitoring the trade-offs among flow, inventory, and quality.

Data

The times for the activities involved in the entire process were modelled using the global variables as follows:

Global Variable name	Value
PrepTimeClient1	13
RoughMachiningClient1	7
CNCTimeClient1	8
InspectTimeClient1	8
PrepTimeClient2	11
RoughMachiningClient2	5
CNCTimeClient2	6
InspectTimeClient2	6
PrepTimeClient3	8
RoughMachiningClient3	3
CNCTimeClient	4
InspectTimeClient3	4

Table 1: Global Variables of the setup

Client Demand Profiles (2880 h horizon)

- ***Client 1 – Precision Gear Housing***
  Interarrival: 72 h, 40 orders over 2880 h (low volume).
  Nature: Tight tolerances, longest processing times, highest inspection burden.
  Implication: Tends to starve its feeder; without leveling it gets pre-empted by higher-volume clients. Use Heijunka to guarantee regular slots and avoid late orders.

- ***Client 2 – Pump Impeller***
  Interarrival: 48 h, 60 orders (medium volume).
  Nature: Moderate complexity and cycle time.

Implication: Acts as the "middle load" that can smooth capacity between C1 and C3; benefits from balanced pull rules (e.g., round-robin with eligibility or pending–in-progress).

- ***Client 3 – Shaft Brackets***
  Interarrival: 28.8 h, 100 orders (high volume).
  Nature: Shortest cycles, simpler geometry, lower defect risk.
  Implication: Can flood the system if not leveled; without FG targets or WIP caps it will dominate throughput and inflate WIP.

Model 1 - Heijunka model

Set-up

The Heijunka model levels three inbound material streams—bars, pre-cast shapes, and sheet-metal—each entering via Queue1, Queue2, and Queue3 to the Preparation station, where a pull rule selects the next item to balance the mix. All parts then pass through Preparation, Queue4, Rough Machining, Queue5, and the CNC station. After CNC, the flow branches into three client lanes: Client 1 uses Queue6 followed by Client1_inspect and the Precision_Gear_Housing finishing step; Client 2 uses Queue7 followed by Client2_inspect; Client 3 uses Queue8 followed by Client3_inspect and the Bracket finishing step. Client demand is modeled with dedicated Order_Queues driven by interarrival times of 72 hours for Client 1, 48 hours for Client 2, and 28.8 hours for Client 3. Completed units are posted to Client-specific "Order_fulfilled" buffers and then consolidated at Orders_Completed.

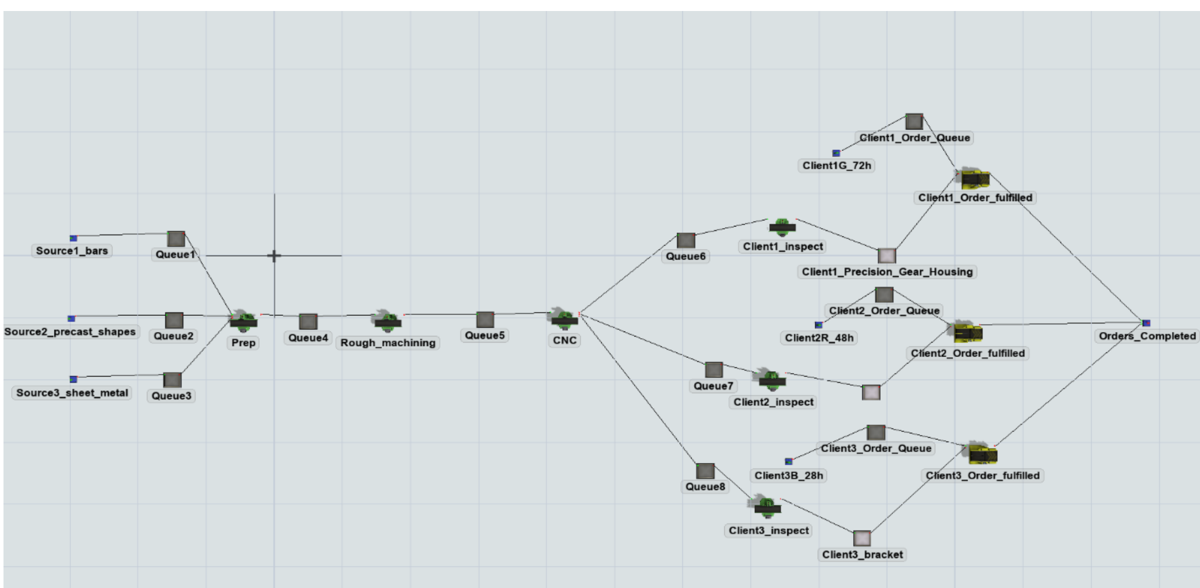


Figure 2:Heijunka model

Pull Strategy

The pull strategy is implemented at the input port of the Prep processor. The Prep pull strategy levels production by selecting the next item for the client whose finished goods (FG) buffer is currently smallest, if client's feeder queue has stock. It first reads FG counts for the three clients (gear housing, impeller, bracket) and the contents of their upstream feeders (Queue1–Queue3). It then adds a small, client-specific random offset to the FG counts to break ties, with the smallest noise on the bracket line and the largest on the gear-housing line, which implements the stated tie priority: blue first, then red, then green. If a feeder is empty, that client is temporarily excluded by assigning a very large surrogate value. Finally, it chooses the minimum adjusted FG and returns the corresponding input-port index (1, 2, or 3) so Prep pulls from the matched feeder.

Prep - Pull From Port

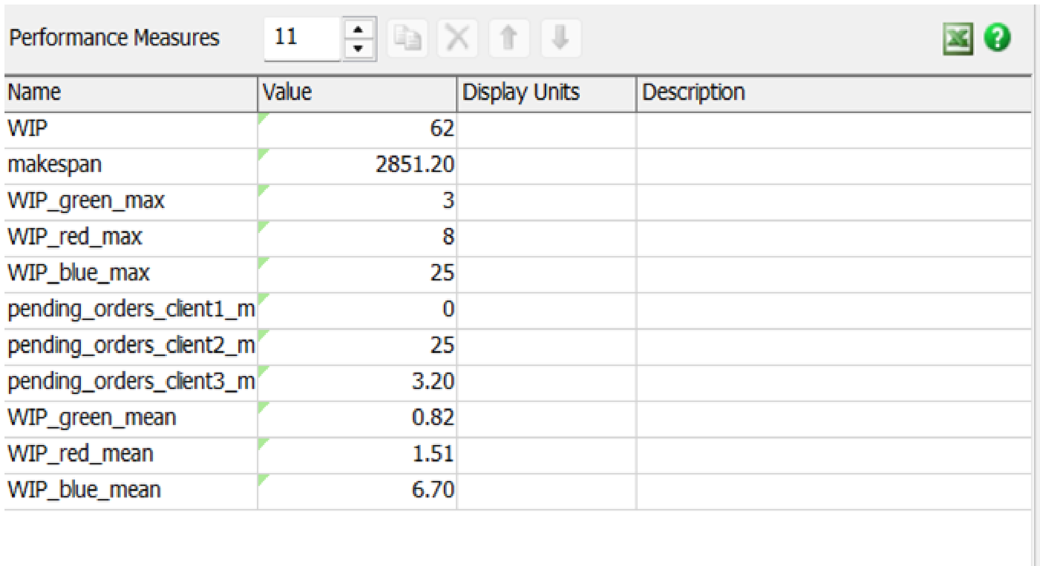
```
Object current = ownerobject(c);
/**Customized*/

double BG=model(). find("Client1_Precision_Gear_Housing"). subnodes. length;
double BR=model(). find("Client2_pump_impeller"). subnodes. length;
double BB=model(). find("Client3_bracket"). subnodes. length;

print ("BG", BG);
print ("BR", BR);
print ("BB", BB);

/* Perform a Production Leveling where Assembly "pulls" the type of item (green, red or blue) whose

//Priority in case of a tie shall be as follows: blue, red and green.

double WG=model(). find("Queue1"). subnodes. length;
double WR=model(). find("Queue2"). subnodes. length;
double WB=model(). find("Queue3"). subnodes. length;

double NewBG=BG+uniform(0,0.3,0);
double NewBR=BR+uniform(0,0.2,0);
double NewBB=BB+uniform(0,0.1,0);

if (WG==0) NewBG= 1000;
if (WR==0) NewBR= 1000;
if (WB==0) NewBB= 1000;

print ("NewBG", NewBG);
print ("NewBR", NewBR);
print ("NewBB", NewBB);

Array myArray=[NewBG, NewBR, NewBB];
double min=Math. min(myArray);
int index_min=myArray. indexOf(min);

print ("min", min);
print ("index_min", index_min);

//Processor input port 1 Plates_Welding connects to the green warehouse
//processor input port 2 Plates_Welding connects to the red warehouse
//processor input port 3 Plates_Welding connects to the blue warehouse
```

Apply All Apply OK Cancel

Figure 3 Pull Strategy code at Prep Processor

Performance measures

The performance measures of the Heijunka model is displayed as follows:



Performance Measures 11

Name	Value	Display Units	Description
WIP	62		
makespan	2851.20		
WIP_green_max	3		
WIP_red_max	8		
WIP_blue_max	25		
pending_orders_client1_m	0		
pending_orders_client2_m	25		
pending_orders_client3_m	3.20		
WIP_green_mean	0.82		
WIP_red_mean	1.51		
WIP_blue_mean	6.70		

Figure 4:Performance measures of Heijunka model

Experimentation

The Heijunka model was evaluated using FlexSim's Experimenter over a 2,880-hour simulation horizon with 100 replications to capture stochastic variability. A single scenario was tested using the parameter set shown in the figure: a buffer capacity of 10, two Rough Machining stations, and three CNC machines. These values represent a mid-capacity configuration intended to support leveled mixed-model production.

The experiment provides a statistical baseline for key KPIs:Completed Orders, Lead Time, WIP, and Makespan, against which the Jidoka model will later be compared.

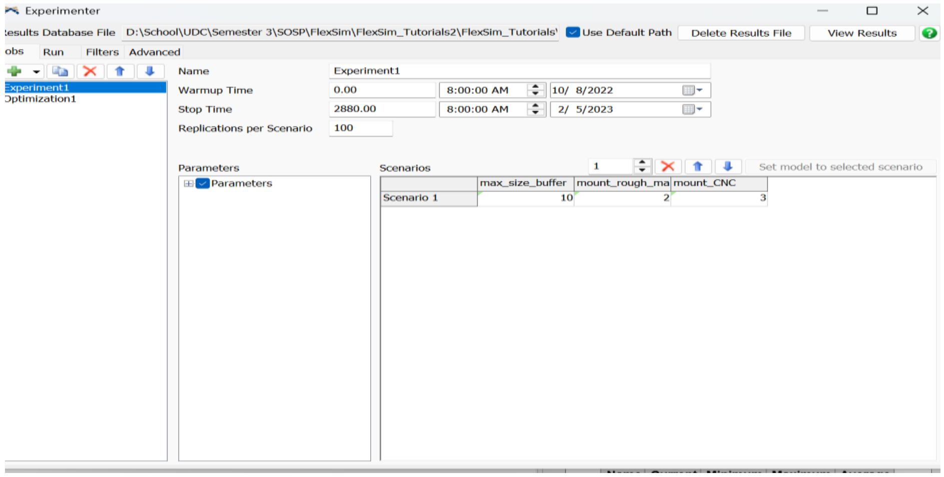


Figure 5 Experimentation set-up

Experimentation Output

Work-in-Process (WIP)

The Heijunka experiment produced highly stable WIP behaviour across the 100 replications. The mean maximum WIP was 61.86 units, with a very narrow 95% confidence interval (±0.07). The minimum and maximum observed values (61 and 62 respectively) show that variability in WIP is almost negligible.

The replication plot confirms this stability: nearly all replications cluster tightly around 62 units, indicating that the production-leveling logic generates a consistent flow pattern regardless of random processing variation. This behaviour is expected in a leveled system, where controlled release and balanced workstation loading limit oscillations in inventory accumulation.

Overall, the Heijunka configuration maintains WIP at a stable and predictable level, demonstrating effective smoothing of the mixed-model production flow.

Figure 6 WIP

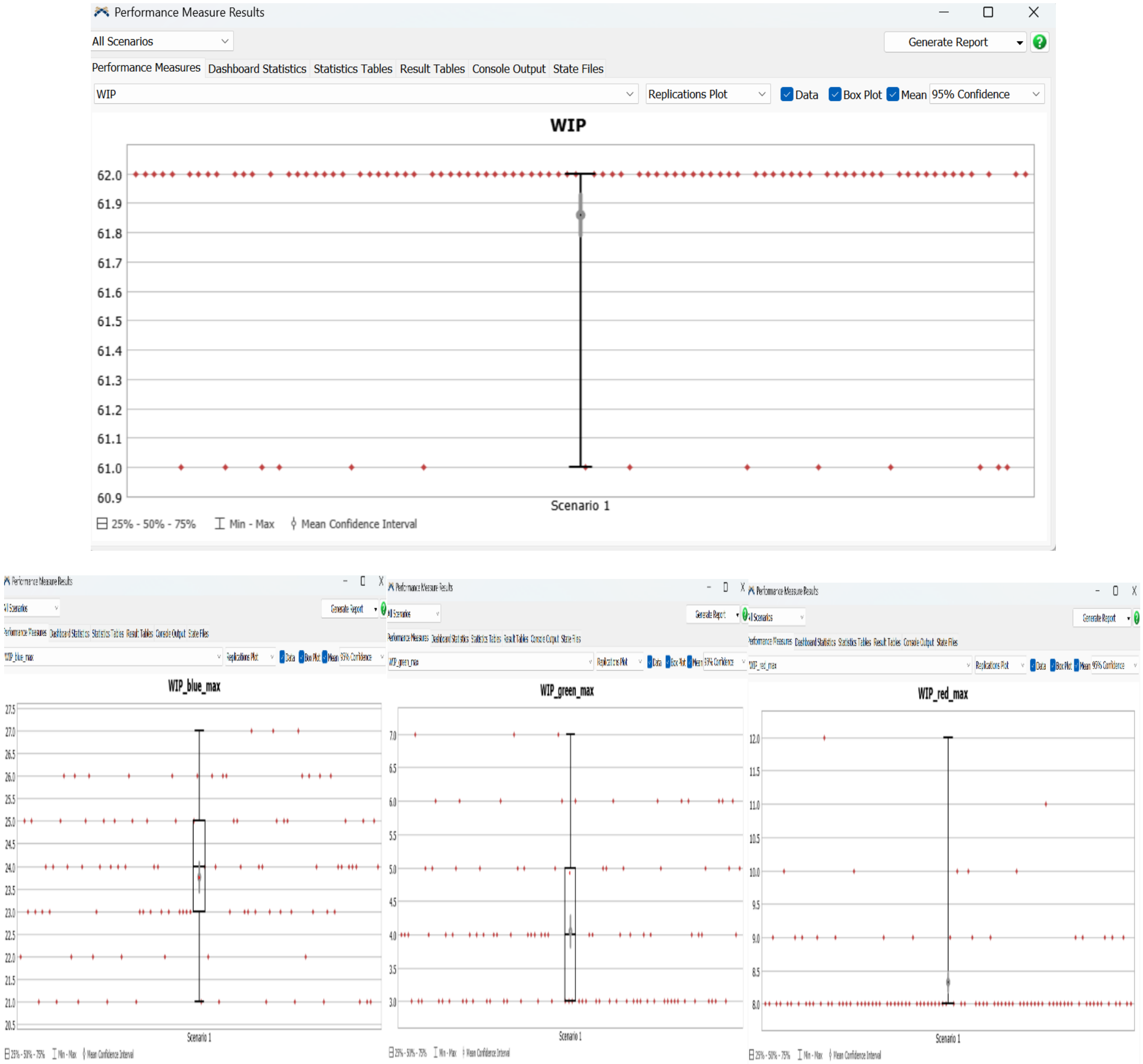


Figure 7 WIP for 3 products

Model 2 – Jidoka model

Set-up

The set-up of the Jidoka model involves the flow of the units from the source, through the processors and the queues without consideration of the clients and the demand. Two defectologies were implemented, one defectology defectology_prep in the Prep processor and the other defectology defectology_cnc in the CNC processor. Two replica models were setup of the same processes and parameters, with one model having the implementation of Jidoka and the other without the Jidoka. The purpose of this setup is to investigate the effect of defectology(errors) on the product along the production line and best course of action for quality control.

Jidoka implemented

Quality control is embedded at the risk-creating stations: defectology_prep at Prep and defectology_cnc at CNC. When a failure/abnormality is detected, the part is diverted to Repairing_Queue → Repairing, then returned to the failing stage (Prep and CNC) for the defect to be corrected. Small in-line queues expose blocking/starvation caused by CNC stops. The result is local containment of defects, lower WIP, and shorter lead time despite short availability losses at CNC.

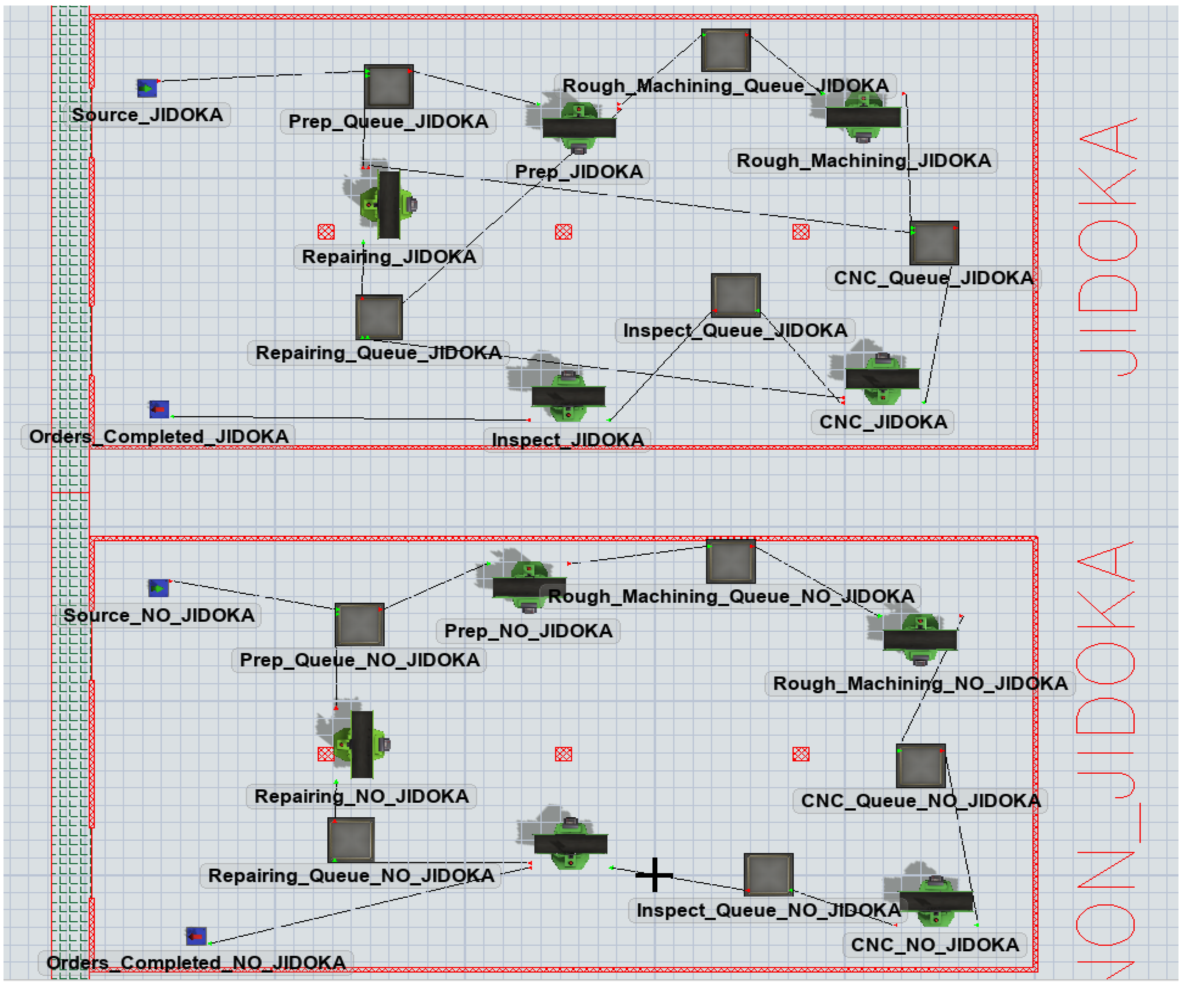


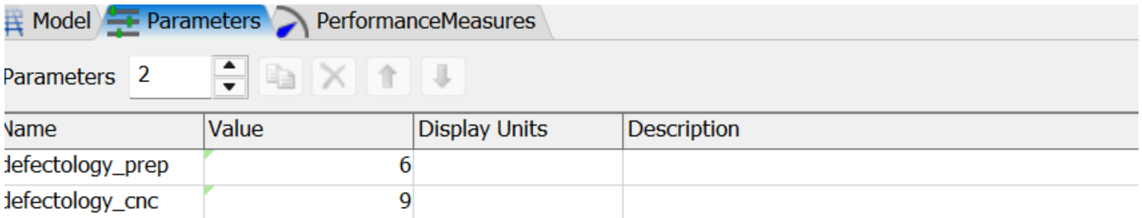
Figure 8:Jidoka model

No Jidoka implemented

The physical sequence replicates the Jidoka line but without autonomation: the CNC does not stop on defect, and detection resolves at Inspect; the failed parts are routed to Repairing_Queue then to Repairing processor and then reintroduced. This will be the baseline highlighting the effect of Jidoka: immediate detection and local rework vs. end-of-line discovery, rendering downtime visible but reducing rework propagation and WIP.

Parameters

The parameters used for the defectologies are shown in the image below:



Name	Value	Display Units	Description
defectology_prep	6		
defectology_cnc	9		

Figure 9 Jidoka parameters

Results and Analysis

Heijunka results

Critical reading of the Heijunka dashboard:

- Orders and completion: "Orders Created and Fulfilled" shows a persistent gap for most of the run (releases outpace completions) with convergence only near the stop time. This indicates the line is capacity-constrained against the aggregate 2:3:5 demand mix; the system drains only at the end (makespan ≈ 2851 h). The Pending Orders panel's steady band confirms continuous arrival pressure throughout the horizon rather than lumpy demand.
- Flow stability and queues: System WIP vs Time forms a classic triangular profile—monotonic build to mid-late run then a controlled drain—which is consistent with a single dominant bottleneck in the common path (Rough/CNC). Average lead time ≈ 1.29 h is low, but the max = 86 h flags occasional starvation or batching effects that the averages hide.
- Per-client behavior: The Clients_Units_Buffers_Staytime plot grows in nearly parallel lines with slopes proportional to demand (green > red > blue), which signals that leveling is generally preserving the product mix. However, the per-client WIP traces reveal late-run stress: Shaft bracket WIP climbs to the 30–40 range, pump impeller spikes to the low teens, while gear housing stays bounded in the 1–6 range. This pattern implies that the pull rule is successfully protecting the low-rate client but allows the high-rate lanes to accumulate work as the horizon approaches—typical when pitch_time and the FG target are static and do not adapt to remaining backlog.

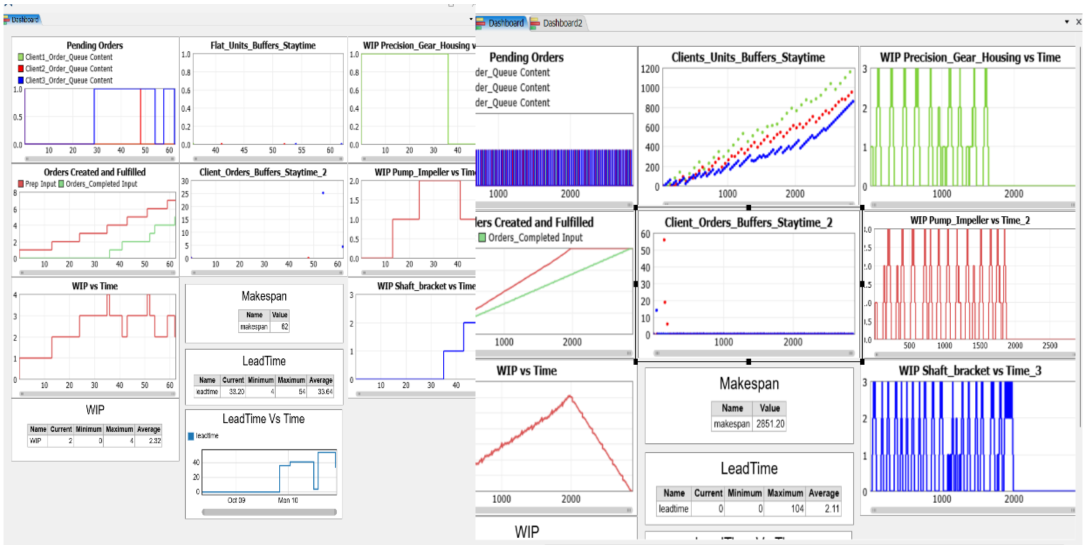


Figure 10:Heijunka Dashboard at initial and final

Heijunka Optimization setup

The Heijunka model was further analysed using FlexSim's OptQuest Optimization module to identify the minimum resource configuration capable of maintaining acceptable system performance. Three decision parameters were selected:

- max_size_buffer – capacity limit for intermediate buffers
- mount_rough_machine – number of Rough Machining stations
- mount_CNC – number of CNC machines

All three were defined as minimization objectives, reflecting the goal of reducing system resources while preserving production performance.

Two operational constraints were imposed:

1. WIP < 62
   Ensures that total Work-in-Process remains below the baseline experimental threshold.
2. Makespan < 2851.2 hours
   Ensures the system completes the required workload within the production horizon.

The optimization was allowed a wall time of 10 seconds and up to 40 iterations. A single replication per scenario was used to speed up the search

The optimization setup for the Heijunka model is shown in the image below:

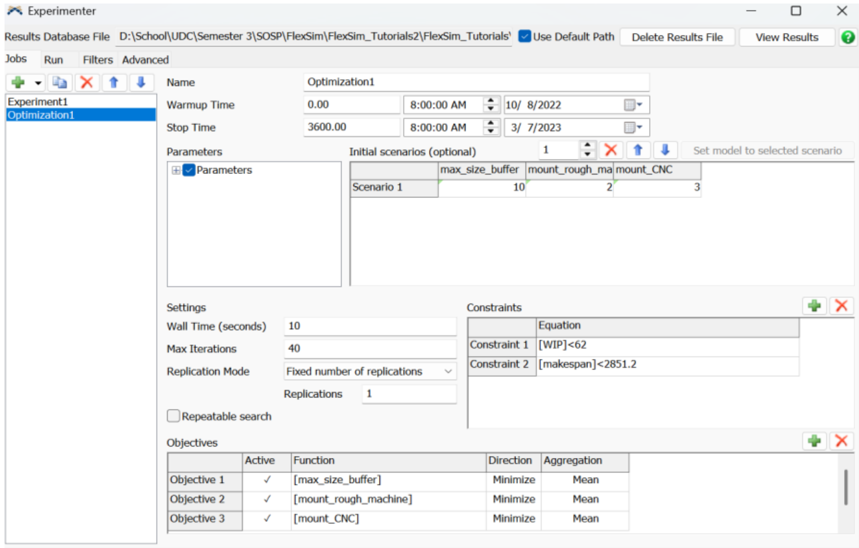


Figure 11:Optimization setup for Heijunka

Heijunka Optimization Output

The best feasible solution found by the optimizer was:

Parameter	Optimal Value
Max_size_buffer	**1**
Mount_rough_machining	**1**
Mount_CNC	**2**

This configuration satisfies both constraints while minimizing resource consumption across all objectives. The scatter plot of iterations shows a wide distribution of trial outcomes, but the optimal scenario consistently appears in the **lower-left region**, representing low resource usage (objectives) and constraint feasibility.

A key observation is that even with significantly reduced resources, particularly shrinking buffer capacity from 10 to 1 and halving Rough Machining capacity, the Heijunka sequencing still maintains system performance within allowable limits. This demonstrates that the leveled production logic inherently stabilizes flow, reducing the need for large buffers and excess machine capacity.

Figure 12 Heijunka optimization solution

Jidoka Results

A table showing the performance measures of the model with both the Jidoka and Non Jidoka simulations are as follows:

Performance Measure	**Values**
Makespan_Jidoka	**1531**
Makespan_No_Jidoka	**1534**
Leadtime_Jidoka	**35.93**
Leadtime_No_Jidoka	**40.31**
WIP_Jidoka	**5**
WIP_No_Jidoka	**6**

Overall, the dashboard shows that the Jidoka line runs cleaner and more stable than the non-Jidoka baseline. Completed orders rise at a comparable rate, while average lead time is slightly lower under Jidoka (≈32.6 vs ≈34.4), indicating faster flow despite occasional autonomous stops. Buffer plots reveal that the Jidoka buffers sit mostly at 0–1 item with short, bounded staytimes, whereas the non-Jidoka case exhibits intermittent spikes, notably at Rough Machining, consistent with defects propagating downstream before being discovered. System-level WIP traces also show a steadier profile for Jidoka, with fewer peaks and quicker recoveries after disturbances. The repairing activity is visible but contained, confirming that catching defects at Prep/CNC and returning parts locally prevents larger queues from forming. In summary, the results support the expected Lean effect: quality at source reduces rework propagation, stabilizes buffers, and improves lead time without hurting overall completions.

Figure 13:Jidoka Dashboard

Jidoka Optimization

The Experimenter was configured as an optimization that searches over the two decision variables defectology_prep and defectology_cnc (the station-specific defect probabilities). The run length is 3600 time units with no warm-up, a single replication, and large limits for wall time and iterations to allow the solver to explore the space. The objectives are set to maximize the two defectology values, while enforcing performance constraints so the line remains acceptable: Makespan_JIDOKA < 1531 and Makespan_NO_JIDOKA < 1534 were treated as separate constraints to provide more insight into the process. In effect, the study identifies the highest defect levels the system can tolerate (with Jidoka and without) without violating the target makespan thresholds, thus quantifying the robustness provided by the Jidoka configuration.

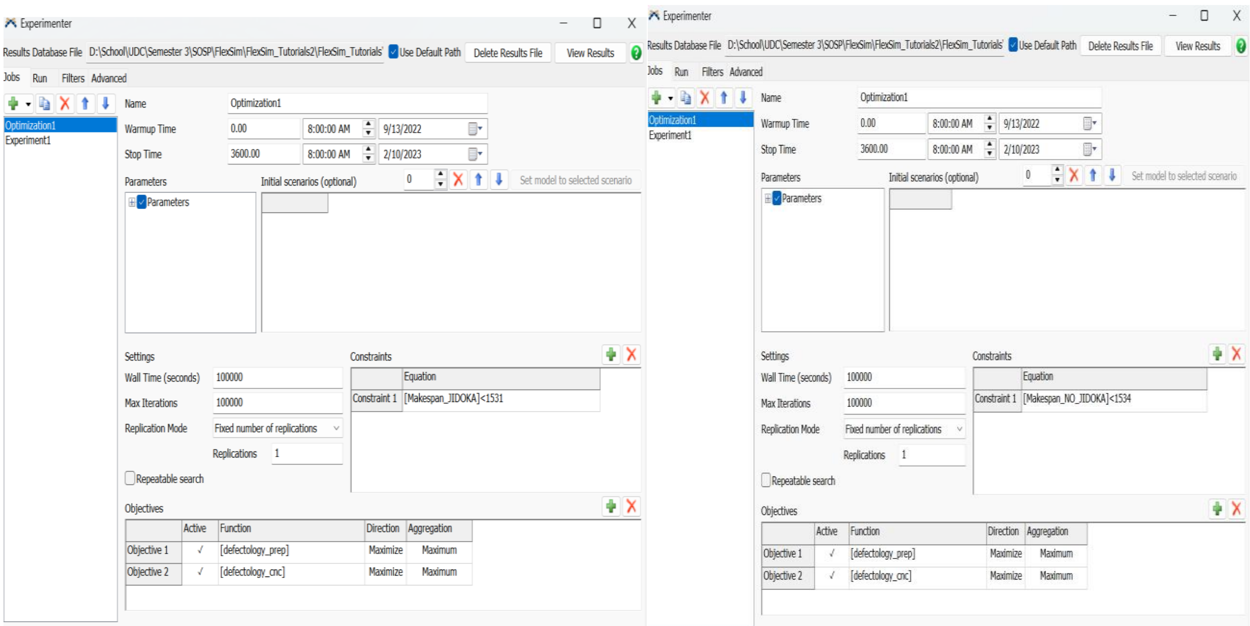


Figure 14:Jidoka Optimization Setup

Jidoka Optimization Output

The optimization explored the maximum defectology levels that the system can tolerate while still meeting strict makespan limits for both the Jidoka and No-Jidoka models. Two defectologies were evaluated as decision variables:

- ***defectology_prep*** → probability of generating a defect in the Preparation stage
- ***defectology_cnc*** → probability of generating a defect in the CNC stage

In both models, the goal was to maximize these defect probabilities while keeping the makespan below the imposed constraint:

- Jidoka constraint: Makespan < 1531 h
- No-Jidoka constraint: Makespan < 1534 h

The defectologies used were implemented at the preparation processor and the CNC processor respectively. Each defectology has a 10% probability of occurring. As discussed in the setup for the optimization, the constraints involving the makespans for both the Jidoka model and the No_Jidoka model were implemented in two distinct optimizations with the results shown in the table:

Constraint 1: Jidoka_makespan<1531

The optimizer determined that the system could tolerate relatively high defect probabilities while remaining under the makespan limit:

Defectology	Model value	Optimal Value
Defectology_prep	**6%**	**10%**
Defectology_cnc	**9%**	**7%**

This result shows that the Jidoka system can handle *full defectology at the first stage* (10%) and a moderately high defect rate at CNC (7%) without violating the makespan constraint. The built-in quality mechanism detecting and repairing defects immediately at the failing stage prevents rework from propagating downstream. This stabilizes flow and keeps the production time within acceptable limits even at elevated defect rates.

Constraint 2: No_Jidoka_makespan<1534

Without Jidoka, the system displays less robustness to defects. The optimized solution converged to:

Defectology	Model value	Optimal Value
Defectology_prep	**6%**	**10%**
Defectology_cnc	**9%**	**10%**

Although both defectologies reached the maximum allowable value (10%), this outcome reflects a key difference in system behaviour. Because defects are not detected until the final inspection, defective items travel through the entire line before being sent to rework. This increases load and WIP across all stations. The slightly higher makespan threshold allowed by the constraint (1534 h instead of 1531 h) provides just enough margin for the No-Jidoka model to absorb this inefficiency.

Interpretation

Overall, the results demonstrate that:

- Jidoka improves process resilience, allowing the system to tolerate high defect rates without exceeding production time limits.
- No-Jidoka is more sensitive to defects, because rework is delayed and affects all downstream stages.
- The optimization confirms that built-in quality (Jidoka) reduces the impact of defects on throughput and makespan, validating the theoretical expectation that early detection and correction prevent wasteful reprocessing and flow instability.

These findings reinforce the practical benefits of Jidoka: faster containment of abnormalities leads to lower makespan variability and better performance under uncertainty.

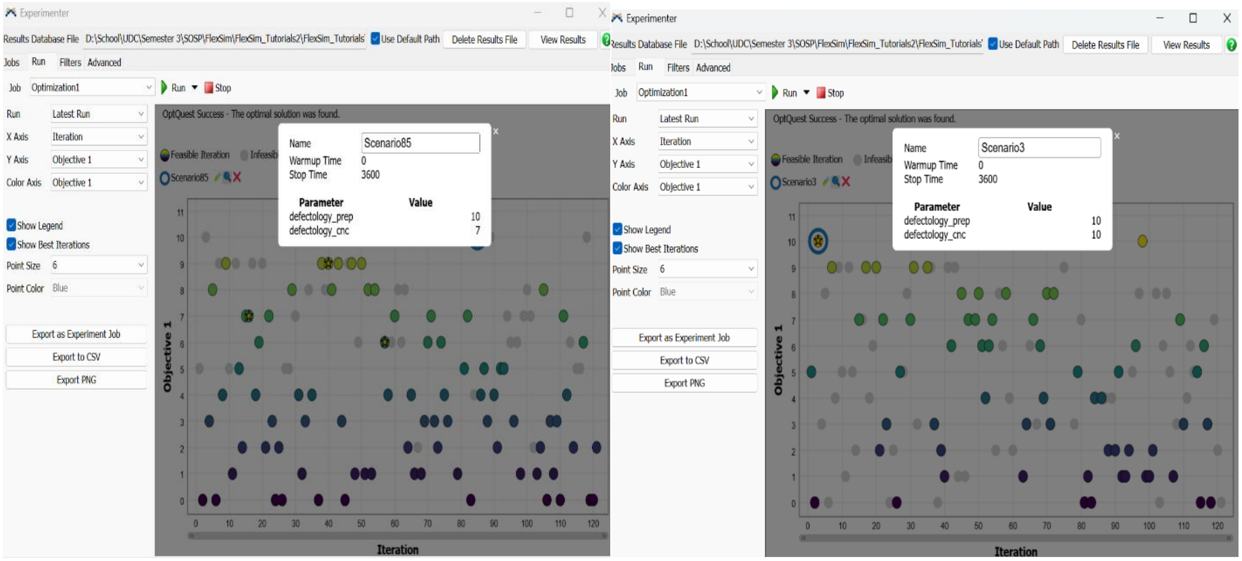


Figure 15:Jidoka Optimation Output

Conclusions

The simulation study demonstrates how Lean Manufacturing principles, specifically Heijunka and Jidoka, improve the performance, stability, and robustness of a CNC machining line operating under mixed-model demand and stochastic quality variation.

- Heijunka successfully leveled production across the three client families, reducing flow variability and maintaining a highly stable WIP profile. Even under random processing variation, WIP remained tightly bound (62 units) with minimal dispersion, confirming effective load balancing. Optimization further showed that the leveled system can operate with greatly reduced buffers and fewer machines while still meeting makespan constraints, highlighting the efficiency gains unlocked through production leveling.

- Jidoka introduced quality-at-source and significantly improved resilience to defects. Immediate detection and local rework prevented defective items from propagating through downstream stages, reducing both WIP and lead time compared to the non-Jidoka baseline. Under equal conditions, the Jidoka line achieved lower lead times (≈36 h vs. 40 h), lower WIP (5 vs. 6), and comparable throughput. Optimization confirmed that the Jidoka system tolerates higher defect rates without violating makespan limits, quantifying the stabilizing effect of autonomation.

Overall, the combined findings show that leveling production and embedding quality at the source fundamentally enhances flow stability, resource efficiency, and robustness in CNC operations

Recommendations

Some recommendations are as follows

1. Adopt Heijunka sequencing to regulate the release of mixed-model demand; this minimizes WIP growth and prevents system overload by high-volume products.
2. Implement Jidoka mechanisms at critical risk-generating stages (e.g., Preparation and CNC) to detect defects early and prevent rework from spreading.
3. Reduce buffer sizes and machine counts cautiously, as optimization results show that a leveled and Jidoka-enabled line can meet production targets with fewer resources, lowering operational costs.
4. Monitor defectology rates, particularly at CNC, since non-Jidoka systems show sensitivity to downstream defect accumulation and may exceed makespan limits under high defect conditions.
5. Use simulation-based optimization prior to system changes, as FlexSim provided reliable insight into the interactions among leveling, defects, and resource utilization.